

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MTH 323 — Past Questions & Materials
by Renance

Course: MTH 323
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 300

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Paper 1 of 1 — MTH 323, 2018

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN, NIGERIA
COLLEGE OF SCIENCE
DEPARTMENT OF MATHEMATICS AND COMPUTER SCIENCE
B.Sc (MATHEMATICS) DEGREE EXAMINATION (2017/2018), FIRST SEMESTER.
MTH 323 (Introduction to Mathematical Modeling-3C), MAY 30, 2018. Duration: 135 mins.

Instructions: Attempt ANY FOUR (4) questions. ALL Questions carries EQUAL marks.
Credit shall be given to students with brevity and concise answers.

Question one

(a) What is a Mathematical Model?

(b) When monitoring the variations of the price of a product, it is observed that a high price for the product above 50 naira attracts more suppliers thus increasing the quantity of items sold. In particular, the price of 200 items decreases the price by 0.1 naira per item. If the current price is 30 naira and 100 items are produced,

- Write down a model that can be used to predict the price and quantity in subsequent years.
- Then find the steady states of the system and explain its behavior.

(c) There are three delivery restaurants near FUPRE called Pizza Paradise, Quick Burger and Noodles Unlimited. They are trying to get as much customers out of 3000 FUPRE undergraduates as possible. A survey conducted showed that 80% of those that ordered pizzas in the past month order pizzas again in the next month, 15% switch to burgers and 5% to noodles. Of those that ordered burgers, 60% order burgers again, 10% order pizzas and 30% order noodles. Of those that ordered noodles, 70% order noodles again, 10% switch to burgers and 20% to pizzas.

Assuming that these tendencies continue and that the number of students remains constant, estimate the long term tendencies.

Question Two

(a) Name three problems that might be modeled mathematically.

(b) Consider the Mathematical Analysis of Affinity Hemodialysis on T-Cell Depletion. Let S be stem cells, T Healthy T cell, T_i infected T cell, v is virus and P is concentration of gp120. Given further that Stem cell S is produced at rate α_1 , from where uninfected T cell is produced at rate α_2 , and increases by recovery of infected cells. The death of bound between healthy cell and, glycoproteins and virus depletes the availability of healthy T cells. Uninfected cells can become infected by virus at a rate α_3 to generate infected cell cells T_i , the infected cell is depleted by virus death or death of the infected cell as well as recovery due to treatment. The viral load is produced from infected cells and bounded to uninfected cell, the virus is reduced by clearance or death at rate β_3 . Glycoproteins 120 is enhanced by viral concentration and depleted by death or clearance at a rate β_5 .

Using the above information, formulate the relevant mathematical equations describing the dynamics of affinity hemodialysis of T-Cells depletion.

(c) Distinguish between Empirical and Theoretical models.

Question Three

(a) List the steps involve in Mathematical model

(b) If there are 70 coins, 10 kobo and 5 kobo, valuing 50 naira, how much of each are there?

(c) A stone is thrown with horizontal velocity 40 ft/sec and vertical (upward) velocity 20 ft/sec from a narrow bridge which is 200 ft above the water.

- How much time elapses before the stone hits the water?
- What is the vertical velocity of the stone just before it hits the water?
- At what horizontal distance from the bridge does the stone hit the water?

$10x + 5y = 5000$

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